Enterprise Digital Forensics & Endpoint Monitoring Platform

1. Overview

This document provides a comprehensive technical overview of an enterprise-grade Digital Forensics and Endpoint Monitoring Platform designed for Windows, macOS, and Linux environments. The platform delivers legitimate, compliant, and security-focused forensic capabilities tailored for enterprise IT, SOC analysts, incident responders, and compliance teams.

The system offers continuous endpoint visibility while maintaining transparency, auditability, and compliance with modern legal, ethical, and enterprise governance standards.

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2. System Objectives

- Provide organizations with a reliable forensic and monitoring tool.
- Support digital investigations, compliance review, and incident response.
- Deliver cross-platform endpoint visibility.
- Ensure all data is collected with proper authorization and logged for audit purposes.
- Supply secure storage, encryption, and tamper-proof evidence handling.
- Support subscription-based licensing for 3, 6, and 12 month cycles.

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3. Core Features

3.1 USB Device Monitoring

Tracks USB device insertions, removals, and file transfers. - Device identification (Vendor ID, Product ID, Serial Number) - File transfer logs (file names, hashes, size, path) - Timeline of events for chain-of-custody

3.2 Screenshot Collection

Allows authorized screenshot capture under enterprise policies. - Configurable screenshot intervals - Secure storage with timestamps - Full audit trail for each capture event

3.3 Screen Recording

Offers optional screen recording functionality. - Administratively controlled recording sessions - Local encryption at rest - Segment-based video recording for optimized storage

3.4 Internet Connection Monitoring

Monitors outbound and inbound network connections. - Active connections (IP, Port, Protocol) - DNS queries and resolution logs - Network events correlated to running processes

3.5 Browser History Collection

Gathers browser activity for incident review. - Chrome/Edge/Firefox/Safari history extraction - URL lists, timestamps, visit count - Optional keyword search for investigations

3.6 Mobile/External Device Connectivity

Provides logs for phone pairing and data transfers. - Android MTP event logs - iPhone lockdown/pairing logs - Identifies files transmitted between device and workstation

3.7 AirDrop / Nearby Share / Screen Mirroring

Monitors discovery and transfer events. - AirDrop broadcast detection - Nearby Share session logs - Screen Mirroring session start/stop tracking

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4. System Architecture

4.1 Agent Layer

The endpoint agent is responsible for collecting forensic events and enforcing policies. - Runs as a service (Windows Service, macOS LaunchDaemon, Linux systemd) - Collects OS-level events and network telemetry - Written in a memory-safe, cross-platform language - Periodically syncs data to the backend using secure channels

Key Agent Responsibilities: - Event monitoring - Local evidence encryption - Policy enforcement - Tamper detection

4.2 Backend Layer

The backend manages ingestion, storage, indexing, search, and policy distribution. - REST/gRPC APIs - Secure data ingestion - Role-based access control (RBAC) - Centralized evidence repository

4.3 Dashboard Layer

A web-based interface for administrators and analysts. - Real-time event visualization - Case management - Device management - Evidence export (JSON, PDF, CSV)

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5. Security Model

Security and integrity are core components of the system.

5.1 Encryption

- TLS/HTTPS for all communication
- AES-256 for evidence storage
- Secure vault for key management

5.2 Tamper Protection

- Service protection mechanisms
- Integrity checks for binaries
- Logs for unauthorized modification attempts

5.3 Audit Logs

Every action—capture, access, download, or configuration change—is logged.

5.4 Compliance Controls

Designed to align with: - SOC2 - GDPR (with transparent consent configuration) - ISO 27001 - Corporate monitoring policies

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6. Evidence Handling

Evidence integrity is maintained from capture to export.

6.1 Hashing

- SHA-256 hashing for all captured files
- Hash stored alongside metadata

6.2 Timestamping

- UTC timestamps for every event
- Logging at millisecond precision

6.3 Chain of Custody

- Automatically generated event timeline
- Detailed context for every action

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7. Deployment Options

7.1 Single-Tenant (On-Premise)

Ideal for highly regulated environments. - Full control over data storage - Local infrastructure

7.2 Multi-Tenant Cloud

Scalable and managed deployment. - Encrypted cloud evidence storage - Automated updates and scaling

7.3 Hybrid

Backend hosted remotely; agents operate locally.

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8. Subscription & Licensing System

The software operates on a subscription basis with license durations: - 3 months - 6 months - 12 months

Licensing system includes: - Hardware fingerprinting (non-invasive) - Online/offline activation - Auto-renewal through the backend - License validation via heartbeat

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9. Technology Stack

9.1 Agent

- Rust (recommended)
- Optional: Go or C++

9.2 Backend

- Rust or Go
- PostgreSQL for structured data
- MinIO/S3 for media artifacts

9.3 Dashboard

- TypeScript
- React / Next.js
- TailwindCSS
- WebSockets for real-time updates

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10. Roadmap

Phase 1 – Foundation

- Core agent development
- USB monitoring
- Network logs
- Dashboard prototype

Phase 2 – Forensics Enhancements

- Screenshot system
- Browser history module
- Device pairing logs

Phase 3 – Advanced Monitoring

- Screen recording
- AirDrop/Nearby Share monitoring
- Real-time alert engine

Phase 4 – Enterprise Hardening

- Subscription management
- RBAC & user groups
- Compliance auditing

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11. Conclusion

This platform is designed to serve as a high-integrity, enterprise-grade digital forensics and endpoint visibility suite. Through secure architecture, transparent monitoring, and compliance-focused design, it delivers legitimate investigative capabilities for modern cybersecurity teams.

Further modules and enhancements can be built as the platform evolves, expanding into full EDR, threat hunting, or automated incident response capabilities.